



Constellation.

Clinton Power Station
8401 Power Road
Clinton, IL 61727

U-604762

10 CFR 50.73

March 31, 2023

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Licensee Event Report 2023-001-00

Enclosed is Licensee Event Report (LER) 2023-001-00: Grid Fault Results in Switchyard Breaker Trips and Reactor Scram. This report is being submitted in accordance with the requirements of 10 CFR 50.73.

There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. David Livingston, Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

Thomas D. Chalmers
Site Vice President
Clinton Power Station

Attachment: Licensee Event Report 2023-001-00

cc:

Regional Administrator - Region III
NRC Senior Resident Inspector - Clinton Power Station
Office of Nuclear Facility Safety - Illinois Emergency Management Agency

IE22
NRR



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Clinton Power Station, Unit 1

☒ 050
☐ 052

2. Docket Number

00461

3. Page

1 OF 3

4. Title

Grid Fault Results in Switchyard Breaker Trips and Reactor Scram

5. Event Date

Month	Day	Year
01	31	2023

6. LER Number

Year	Sequential Number	Revision No.
2023	- 001 -	00

7. Report Date

Month	Day	Year
03	31	2023

8. Other Facilities Involved

Facility Name	☐ 050	Docket Number
Facility Name	☐ 052	Docket Number

9. Operating Mode

Mode 1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☐ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☐ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

David Livingston

Phone Number (Include area code)

217-937-2800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
E	FK	BRK	GE	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 1/31/23, at 2343 CST, while operating at approximately 100% power, a phase-to-phase fault occurred on a 345 KV transmission line approximately 16 miles from Clinton Power Station (CPS). Gas Circuit Breaker Generator Output Breakers (GCBs) 4510 and 4514 tripped open to clear the fault. GCB 4510 failed to trip open properly, resulting in actuation of the GCB 4510 breaker failure protective relay, tripping open GCB 4506 and disconnecting the main generator from the grid. This resulted in a turbine trip and automatic reactor scram. The root cause of the failure of GCB 4510 to properly open, and subsequent trip open of GCB 4506, was due to corrosion of the GCB 4510 air system charging valve. Corrective actions taken and planned include: repairs to the 345 KV transmission line involved in the event, developing and implementing an improved Preventive Maintenance (PM) strategy for switchyard breakers, working with the switchyard operator to perform twelve-year breaker overhaul/replacement on affected GCBs/345 KV switchyard breakers, and providing training on effective design attribute review performance. There was no impact to the health and safety of the public or plant personnel from this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME

Clinton Power Station, Unit 1

☒ 050☐ 052**2. DOCKET NUMBER**

00461

3. LER NUMBER

YEAR

SEQUENTIAL
NUMBERREV
NO.

2023

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NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric -- Boiling Water Reactor, 3473 Megawatts Thermal Rated Core Power
Energy Industry Identification System (EIS) codes are identified in text as [XX].

EVENT IDENTIFICATION

Grid Fault Results in Switchyard Breaker Trips and Reactor Scram

A. Plant Operating Conditions Before the Event

Unit: 1

Event Date: January 31, 2023

Event Time: 2343 CST

Mode: 1

Mode Name: Power Operation

Reactor Power: 100

B. Description of Event

On January 31, 2023, at 2343 CST, while operating at approximately 100% power, a phase-to-phase fault occurred on a 345 KV transmission line approximately 16 miles from Clinton Power Station (CPS). In response to the fault, Gas Circuit Breaker Generator Output Breakers (GCBs) 4510 and 4514 [FK:BRK] tripped open to clear the fault. GCB 4510 failed to trip open properly, resulting in actuation of the GCB 4510 breaker failure protective relay, tripping open GCB 4506 [FK:BRK] and disconnecting the main generator [EL:GEN] from the grid. This resulted in a turbine [TA:TRB] trip and automatic reactor [RCT] scram. Operators placed the reactor mode switch in shutdown and completed the scram response actions. All control rods fully inserted.

As noted above, GCB 4514 tripped open successfully. However, breaker failure protective relaying was actuated due to detection of current flow through GCB 4510 after it received a trip open signal. This resulted in tripping open GCB 4506 to clear the fault.

C. Cause of the Event

When a GCB breaker opens, the breaker control circuits energize trip coils. Energizing the trip coils creates a magnetic force that releases a trip latch, allowing spring energy to open an air pilot valve. The pilot valve allows air from the associated air receiver to flow into the breaker's operating piston/operating cylinder. The compressed air forces the operating piston to stroke the operating rod, which is connected to the interrupter. In addition to opening the interrupter, the operating piston compresses the breaker's closing springs as it travels. The operating piston overtravels slightly allowing the breaker closing latch to cycle and capture the operating rod in the open position against closing spring pressure. The breaker will remain open until the closing latch is released by the associated closing coils. In this event, GCB 4510 failed to latch open.

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CONTINUATION SHEET**

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00461

3. LER NUMBER

YEAR	SEQUENTIAL NUMBER	REV NO.
2023	- 001	- 00

NARRATIVE

The root cause of the failure of GCB 4510 to latch open and subsequent trip open of GCB 4506 was determined to be due to corrosion of the GCB 4510 air system charging valve due to expected moisture that was not mitigated because the manufacturer's recommended twelve-year breaker maintenance was not performed.

D. Safety Consequences

The condition described in this LER is reportable under 50.73(a)(2)(iv)(A), any event or condition that resulted in manual or automatic actuation of any of the systems listed in paragraph (a)(2)(iv)(B). All safety systems operated as designed during and following the scram. As a result, there was no impact to the health and safety of the public or plant personnel from this event. In addition, this event does not meet the criteria for a Safety System Functional Failure.

E. Corrective Actions

Corrective actions taken and planned include:

- Grid operator completed necessary repairs to the 345 KV transmission line involved in the event.
- Develop and implement an improved Preventive Maintenance (PM) strategy for switchyard breakers considering failure history, vendor recommendations, industry operating experience, and engineering judgement.
- Work with switchyard operator to perform twelve-year breaker overhaul or replacement on GCBs 4506 and 4510 as specified in vendor manual.
- Work with switchyard operator to perform twelve-year breaker overhaul or replacement on applicable 345 KV switchyard breakers.
- Provide training on effective design attribute review performance to the appropriate population.

F. Previous Similar Occurrences

A review of previous LERs did not identify any events that were similar to the condition described in this LER.

G. Component Failure Data

GCB 4510 is a GE-Hitachi High Voltage Breaker (HVB) SF6 GCB breaker.